

Turn this assignment in on gradescope. All work must be neat and legible. Turn in a "final draft". There should be nothing scratched out and your work should flow generally from left to right and top to bottom.

1. In the text, it says that Newton was able to come up with the series for $\sqrt{1+x}$ using the "standard" root extraction algorithm. Find that algorithm and compute the series using Newton's Method.

Newton adapted the standard long square root algorithm (the pencil-and-paper method for extracting square roots digit-by-digit) to work on power series.

To find $P = \sqrt{1+x}$, write $P = a_0 + a_1x + a_2x^2 + \dots$ and build it term by term.

At each step you have a partial sum p and a remainder $r = (1+x) - p^2$. The next term δ satisfies:

$$(p + \delta)^2 = p^2 + 2p\delta + \underbrace{\delta^2}_{\text{higher order}} \approx 1 + x$$

So $\delta \approx \frac{r}{2p}$, and specifically $\delta = \frac{\text{leading term of } r}{2 \cdot \text{leading term of } p}$.

Then update: $r \leftarrow r - [(p + \delta)^2 - p^2] = r - 2p\delta - \delta^2$.

The first few steps are as follows:

Step 1: $p = 1$ gives $p^2 = 1$. Remainder: $r = (1+x) - 1 = x$.

Step 2: $\delta = \frac{x}{2 \cdot 1} = \frac{x}{2}$. Update $p = 1 + \frac{x}{2}$.

$$r \leftarrow x - 2(1) \left(\frac{x}{2}\right) - \left(\frac{x}{2}\right)^2 = x - x - \frac{x^2}{4} = -\frac{x^2}{4}$$

Step 3: $\delta = \frac{-x^2/4}{2 \cdot 1} = -\frac{x^2}{8}$. Update $p = 1 + \frac{x}{2} - \frac{x^2}{8}$.

If we continue this process, we get the result:

$$\sqrt{1+x} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \dots$$

This matches the binomial series $(1+x)^{1/2}$ with the general term:

$$\binom{1/2}{n} x^n = \frac{\frac{1}{2} \left(\frac{1}{2} - 1\right) \cdots \left(\frac{1}{2} - n + 1\right)}{n!} x^n$$

2. Find the curvature of the ellipse $x^2 + 4y^2 = 1$ by using Newton's procedure.

Take fluxions of $x^2 + 4y^2 = 1$ with $\dot{x} = 1$:

$$2x + 8y\dot{y} = 0 \implies z = \dot{y} = -\frac{x}{4y}.$$

Take fluxions again (using $\dot{x} = 1$, $\dot{y} = z$):

$$2 + 8z^2 + 8y\dot{z} = 0 \implies \dot{z} = -\frac{1 + 4z^2}{4y}.$$

Substituting $z = -x/4y$, so $4z^2 = x^2/4y^2$:

$$1 + 4z^2 = \frac{4y^2 + x^2}{4y^2} = \frac{1}{4y^2}$$

where the last step uses $x^2 + 4y^2 = 1$. Thus $\dot{z} = -\frac{1}{16y^3}$.

Substituting $z^2 = x^2/16y^2$ and $x^2 = 1 - 4y^2$:

$$1 + z^2 = \frac{16y^2 + x^2}{16y^2} = \frac{12y^2 + 1}{16y^2}.$$

Newton's formula then gives:

$$DC = \frac{(1 + z^2)^{\frac{3}{2}}}{\dot{z}} = \frac{\left(\frac{12y^2 + 1}{16y^2}\right)^{3/2}}{\frac{1}{16y^3}} = \frac{(12y^2 + 1)^{3/2}}{64y^3} \cdot 16y^3 = \frac{(12y^2 + 1)^{3/2}}{4}.$$

3. Derive the product rule using differentials.

Let u and v be two quantities depending on x . Consider their product uv at x and at $x + dx$:

$$d(uv) = (u + du)(v + dv) - uv.$$

Expanding:

$$d(uv) = uv + u dv + v du + du dv - uv.$$

The term $du dv$ is the product of two infinitesimals, and so is very small compared to the remaining terms. If we discard it:

$$d(uv) = u dv + v du.$$

4. Using Leibniz's Method, find the tangent to $x^2 + xy + y^2 = 7$ at the point $(1, 2)$.

Take the differential of both sides:

$$d(x^2) + d(xy) + d(y^2) = d(7).$$

Applying the rules for differentials (using the product rule for $d(xy)$):

$$2xdx + xdy + ydx + 2ydy = 0.$$

Collect terms in dx and dy :

$$(2x + y)dx + (x + 2y)dy = 0.$$

Solving for the ratio dy/dx (the slope of the tangent):

$$\frac{dy}{dx} = -\frac{2x + y}{x + 2y}.$$

At the point $(1, 2)$:

$$\left. \frac{dy}{dx} \right|_{(1,2)} = -\frac{2(1) + 2}{1 + 2(2)} = -\frac{4}{5}.$$

The tangent line through $(1, 2)$ with slope $-4/5$:

$$y - 2 = -\frac{4}{5}(x - 1).$$